

Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Project Title

MediLink – Smart Doctor Appointment & Healthcare Management System.

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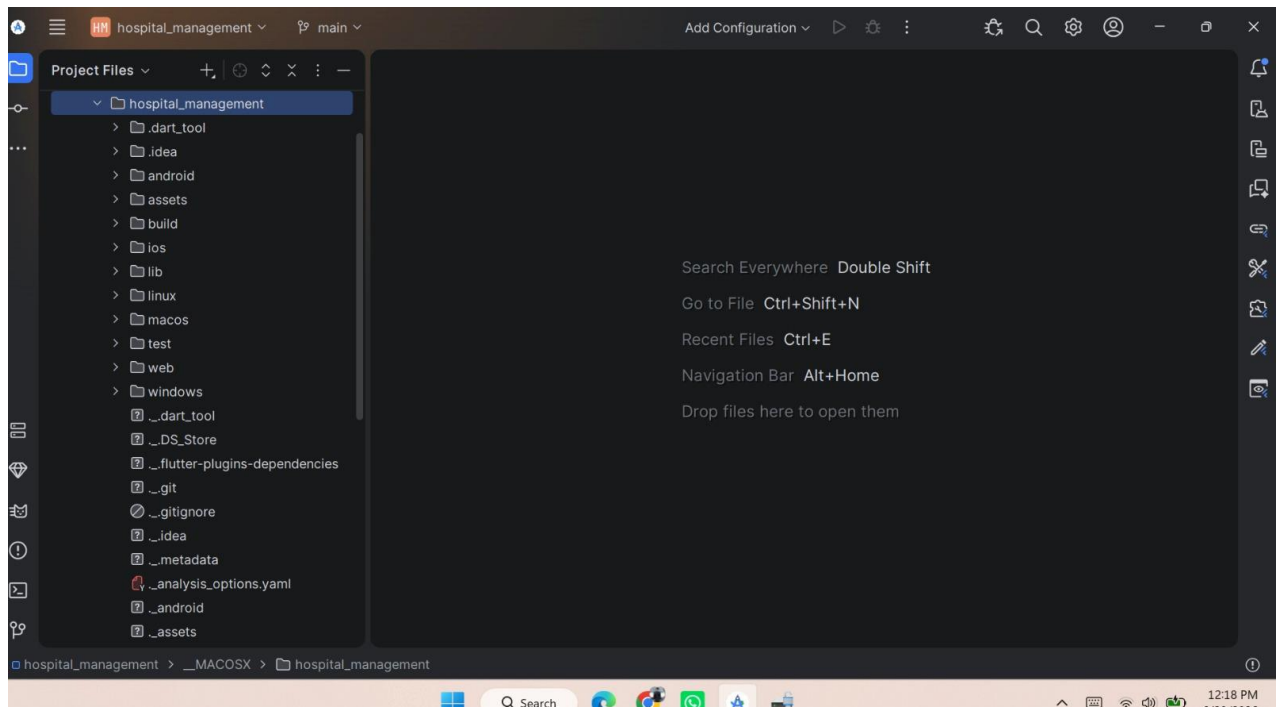
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Frontend Technology Stack



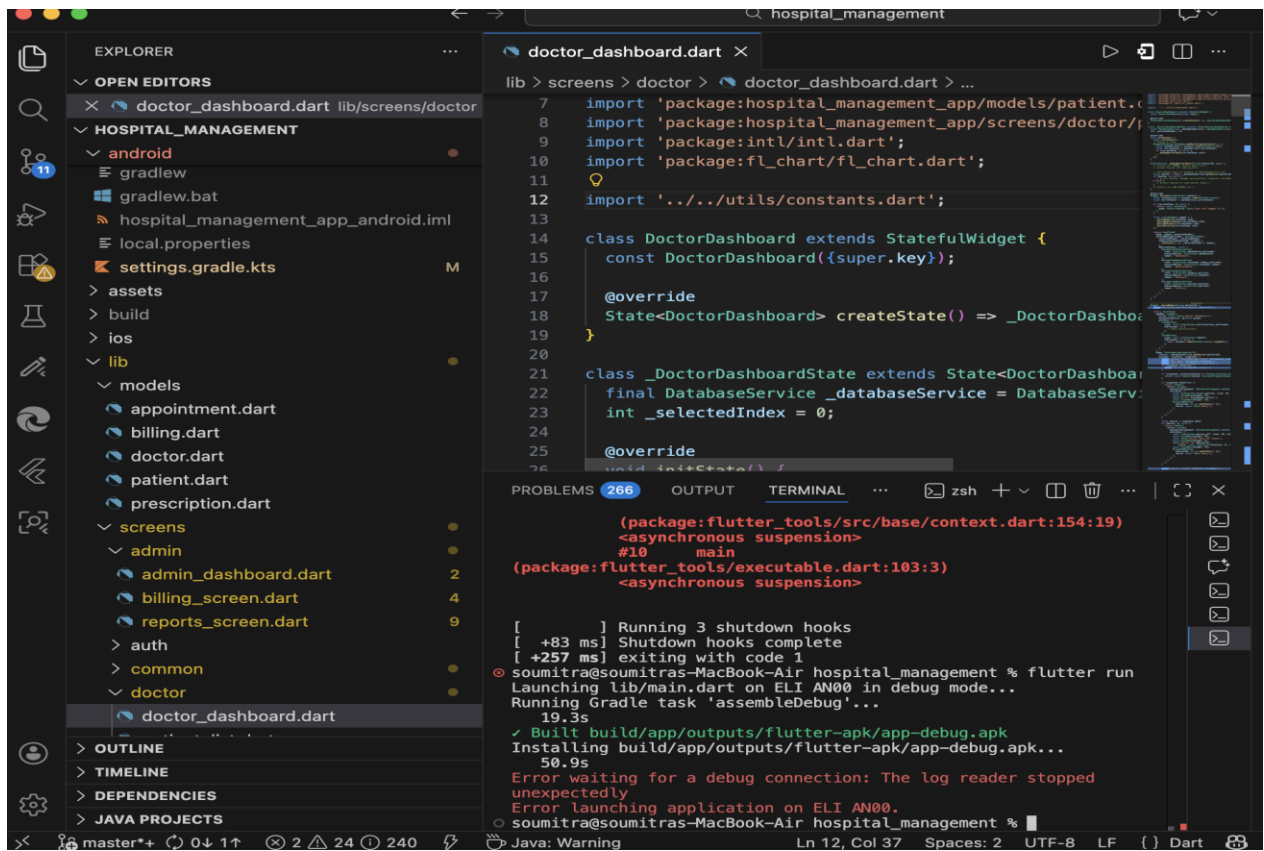
The frontend of the MediLink project is developed using **Flutter**, a cross-platform UI toolkit that enables the development of mobile applications for both **Android and iOS** from a single codebase. The programming language used is **Dart**, which provides efficient and flexible UI development.

Flutter is used to build a modern, responsive, and user-friendly interface for the MediLink healthcare system, ensuring smooth interaction for both patients and doctors. The application follows Material Design principles to maintain a clean and consistent user experience across all screens.

Key Frontend Features:

- Developed using **Flutter SDK** for cross-platform mobile application development
- Implemented with **Dart programming language** for UI logic and functionality
- Clean and modern **Material Design UI**
- Fully **responsive layout** supporting different screen sizes
- Efficient **state management** for smooth data handling
- Seamless integration with **Firebase services** (Authentication, Firestore, Cloud Messaging, Storage)

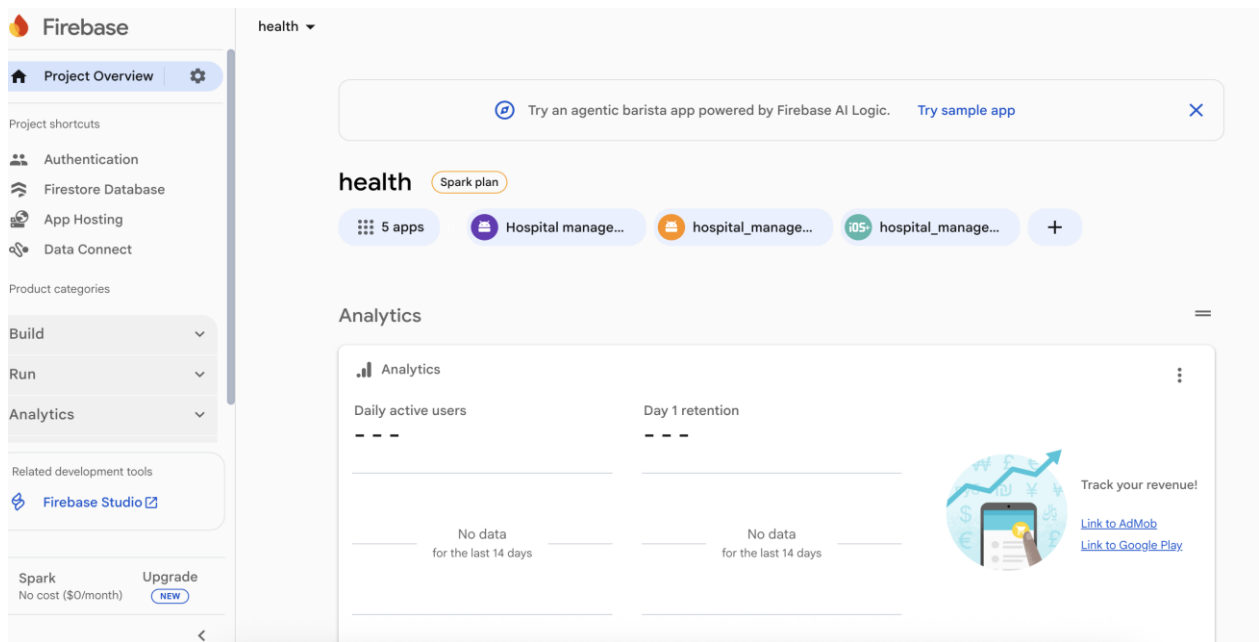
Backend Technology Stack



MediLink is a Smart Doctor Appointment and Healthcare Management System developed using Flutter for the frontend and Firebase as the complete backend solution. Firebase Authentication is used for secure user login and registration, while Cloud Firestore manages real-time data such as patient records, doctor profiles, and appointments. Firebase Cloud Messaging (FCM) handles push notifications for appointment updates and reminders. This system ensures fast, scalable, and secure healthcare service management without requiring a traditional server.

- Firebase (Complete Backend Solution)
- Firebase Authentication (User Login & Registration)
- Cloud Firestore (Real-time Database)
- Firebase Cloud Functions (Server-side Logic)
- Firebase Cloud Messaging (Push Notifications)
- Firebase Storage (File/Image Storage)

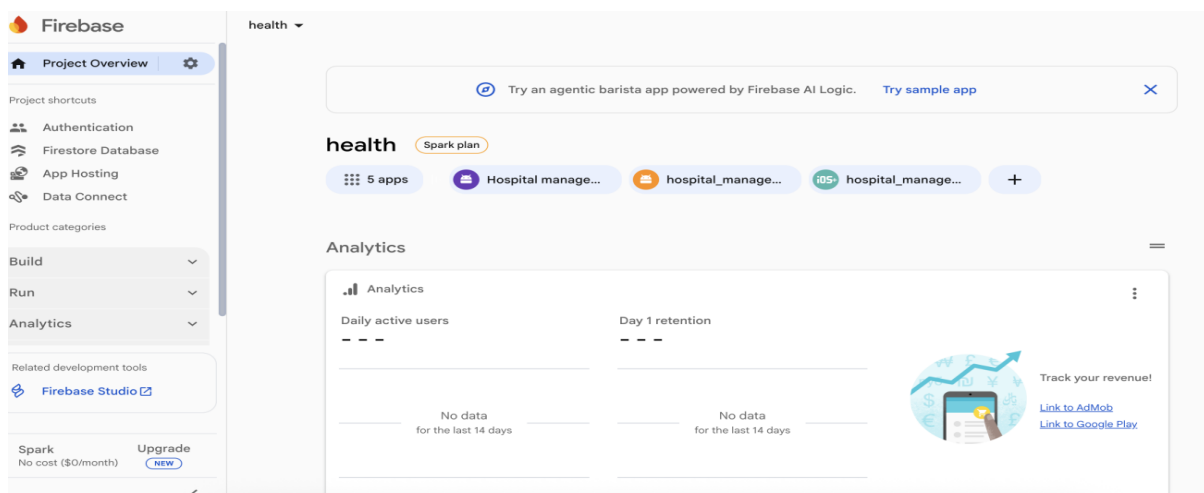
Firestore System



Recent systems leverage cloud platforms to store and manage healthcare data. These solutions offer benefits such as scalability, high availability, and real-time data synchronization. Cloud-based HMS often use backend services to manage authentication, database operations, and file storage. However, some existing implementations lack strong role-based access controls and may not fully address data privacy and compliance requirements.

Cloud Storage

Manages file-based data such as medical reports, prescriptions, and scanned documents. These files are securely stored and linked to the relevant patient records



Authentication Module



The diagram also illustrates how data flows from the application to the backend and is synchronized in real time, ensuring that all users receive updated information instantly. This structured flow ensures efficient system operation, data security, and smooth communication between hospital staff and patients.

The flow diagram illustrates the functional workflow and data movement within the Hospital Management App developed using Flutter and Firebase. It provides a clear view of how different user roles interact with the system and how information is processed and stored securely through the backend infrastructure.

Login Screen



Figure : Login Screen

Register Screen

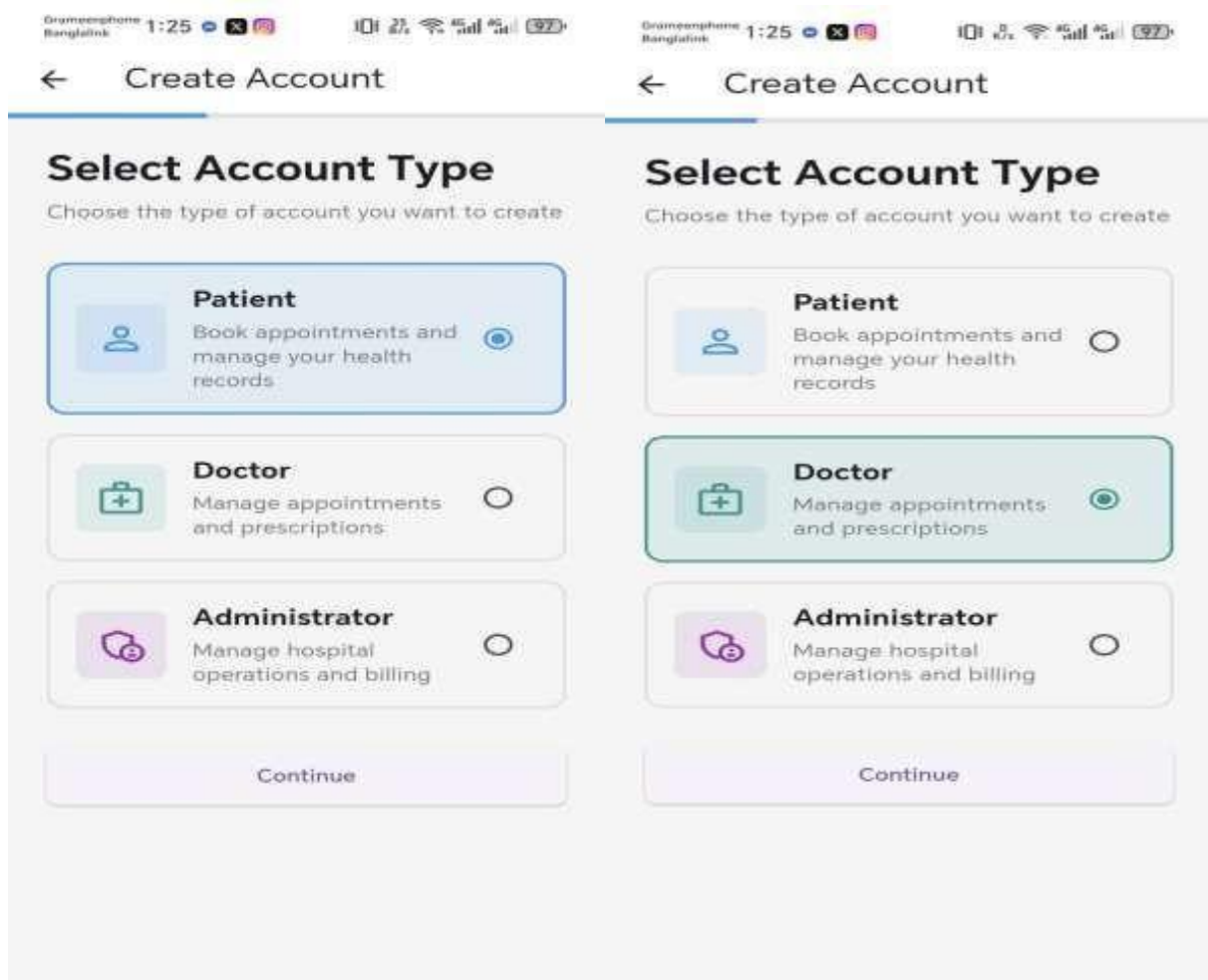


Figure : Register Screen

The register screen enables new users to create an account in the MediLink App by providing essential details such as name, email address, contact information, and password. The registration process is integrated with Firebase Authentication to ensure secure and reliable user account creation. The screen includes basic input validation to check for missing or incorrect information and provides feedback messages to guide users during the registration process. After successful registration, user data is stored in the database and the user is assigned a role, which determines their access level within the system. This screen plays a key role in maintaining data accuracy, system security, and smooth onboarding of new users.

Patient Dashboard

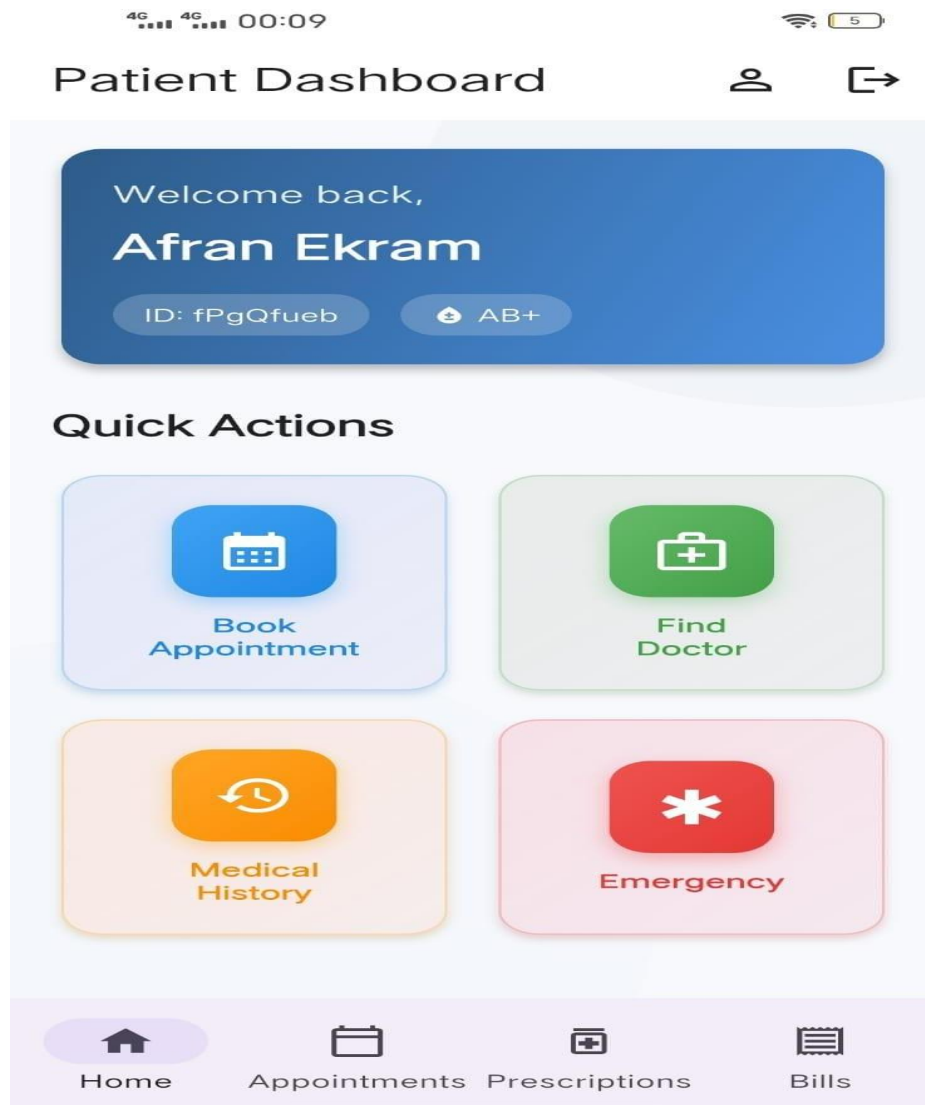


Figure : Patient Dashboard

The patient dashboard serves as the main interface for patients after successful login into the MediLink App. It provides a centralized view of key features such as appointment booking, viewing upcoming and past appointments, accessing medical records, and checking billing information. The dashboard is designed with a clean and user-friendly layout to ensure easy navigation and quick access to important services. Real-time updates from the Firebase backend allow patients to receive instant notifications about appointment confirmations, schedule changes, and system alerts. This screen enhances patient engagement and improves communication between patients and healthcare providers.

Doctor Dashboard

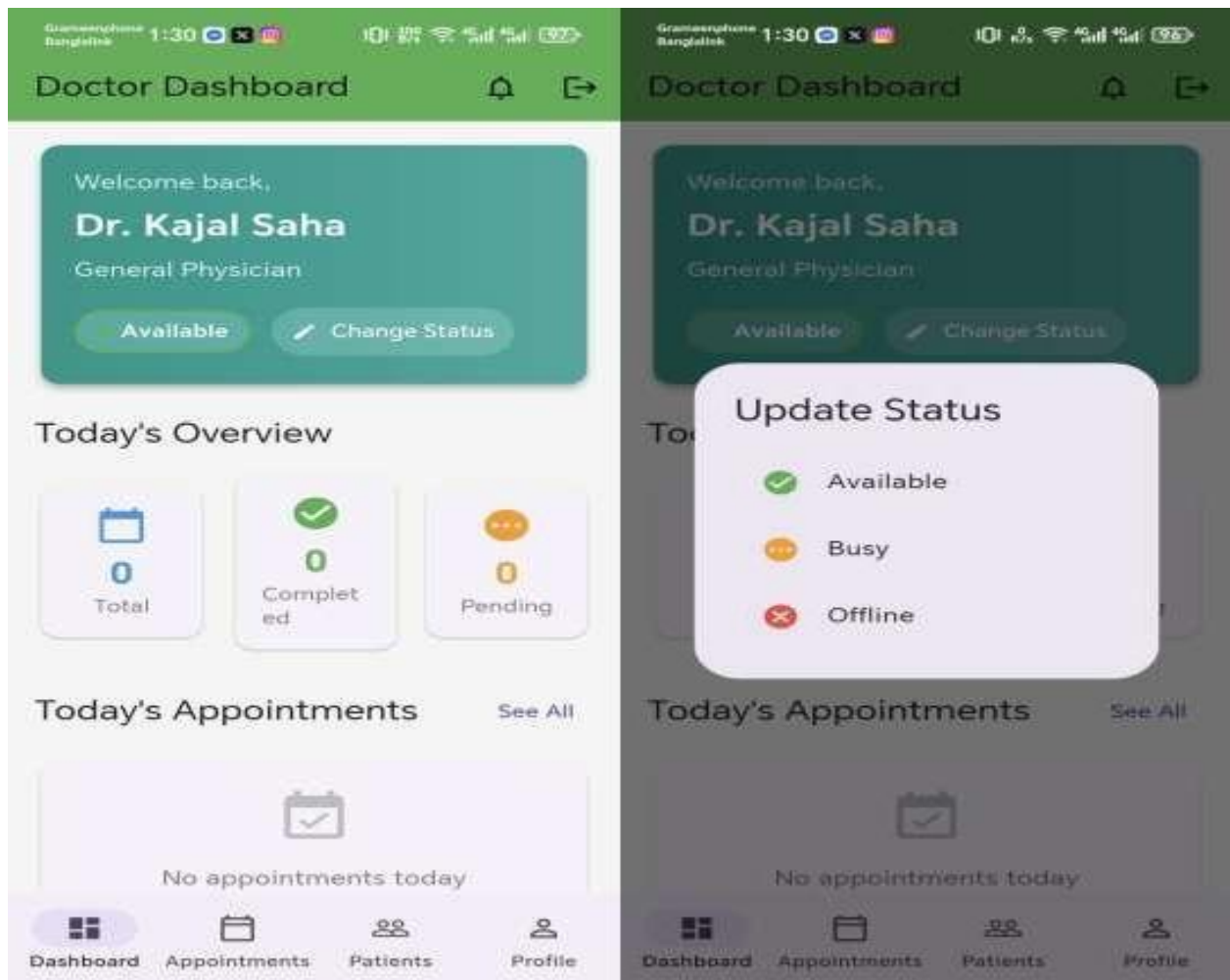


Figure : Doctor Dashboard

The doctor dashboard provides a centralized interface for doctors to manage their daily activities within the MediLink App. It displays scheduled appointments, patient details, and access to medical records in an organized manner. Doctors can view patient histories, update consultation notes, and manage their availability through this screen. Realtime data synchronization via Firebase ensures that any updates made by the doctor are instantly reflected across the system, improving coordination and enhancing the quality of patient care.

Book Appointment

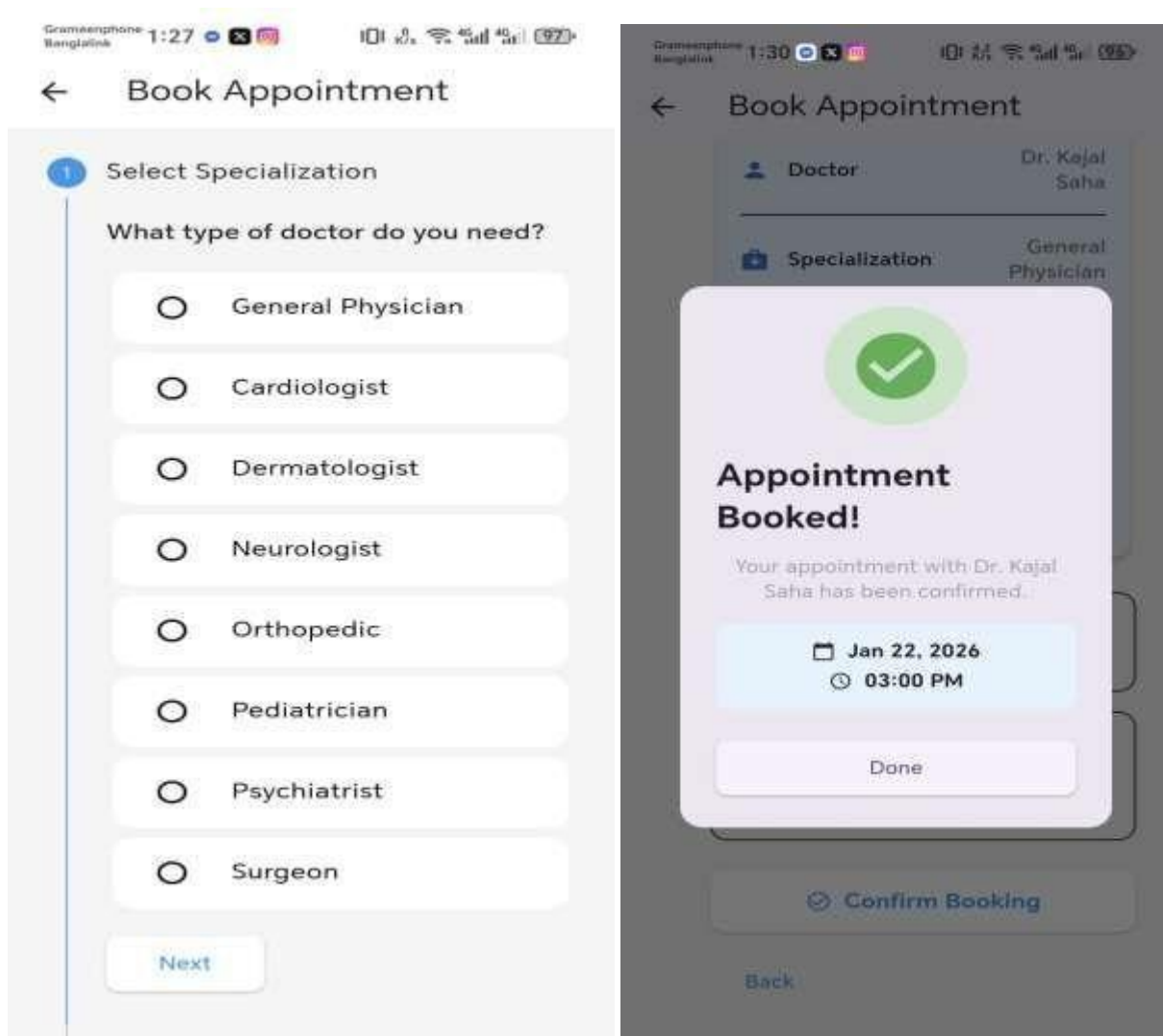


Figure : Book Appointment

The Book Appointment screen allows patients to schedule consultations with doctors through a simple and guided process. Patients can select the desired department, choose an available doctor, and pick a suitable date and time slot based on real-time availability. The system validates the input data and stores the appointment details securely in the Firebase database. Once the booking is successful, confirmation messages and notifications are generated for both the patient and the doctor. This feature reduces manual scheduling efforts, minimizes waiting time, and improves overall coordination within the hospital system.

Logout

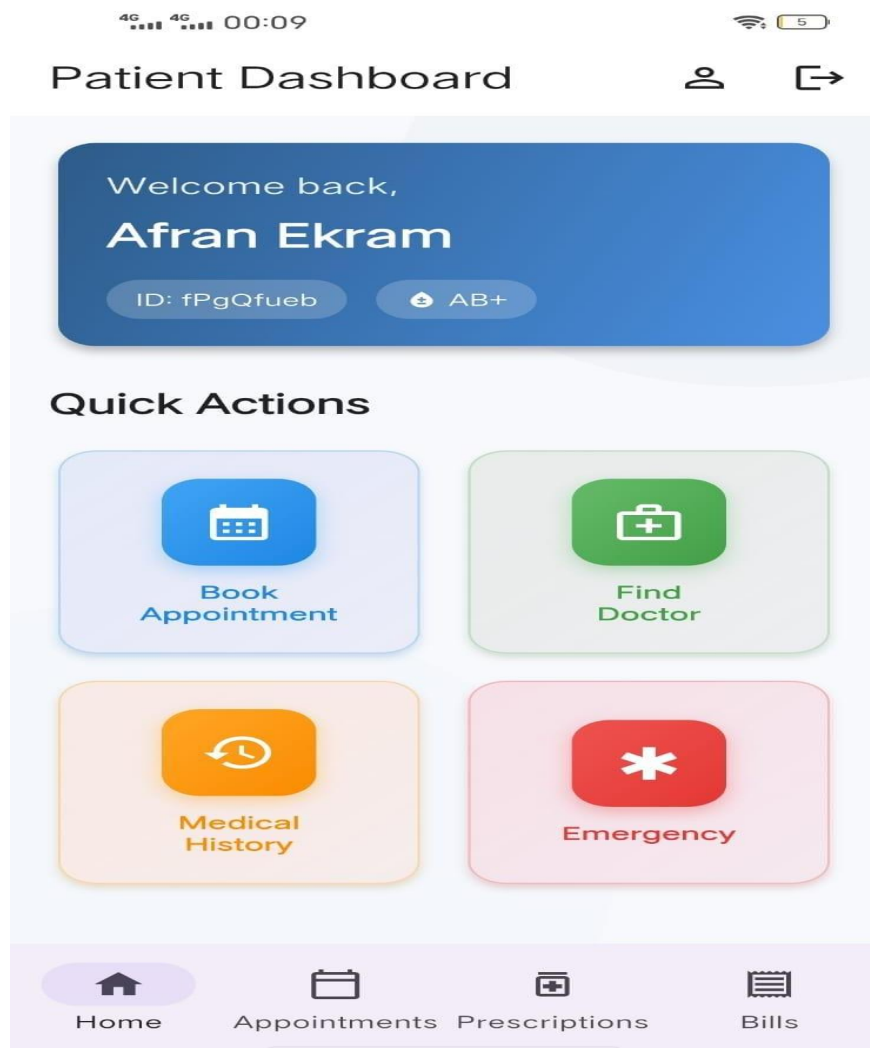
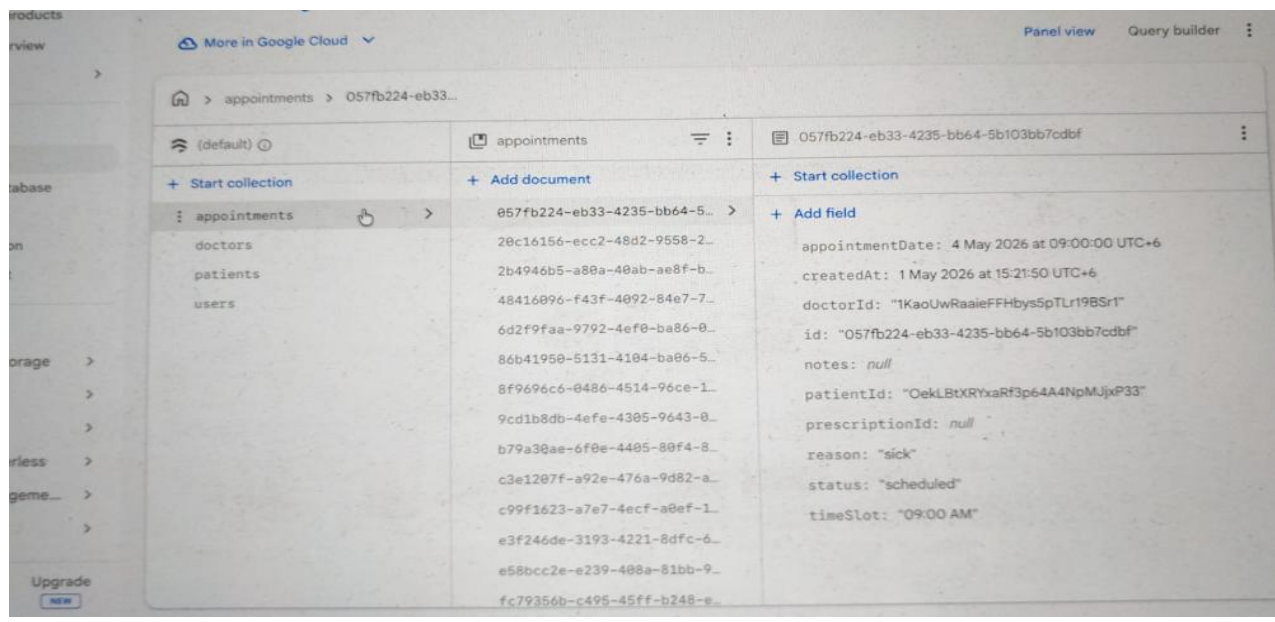


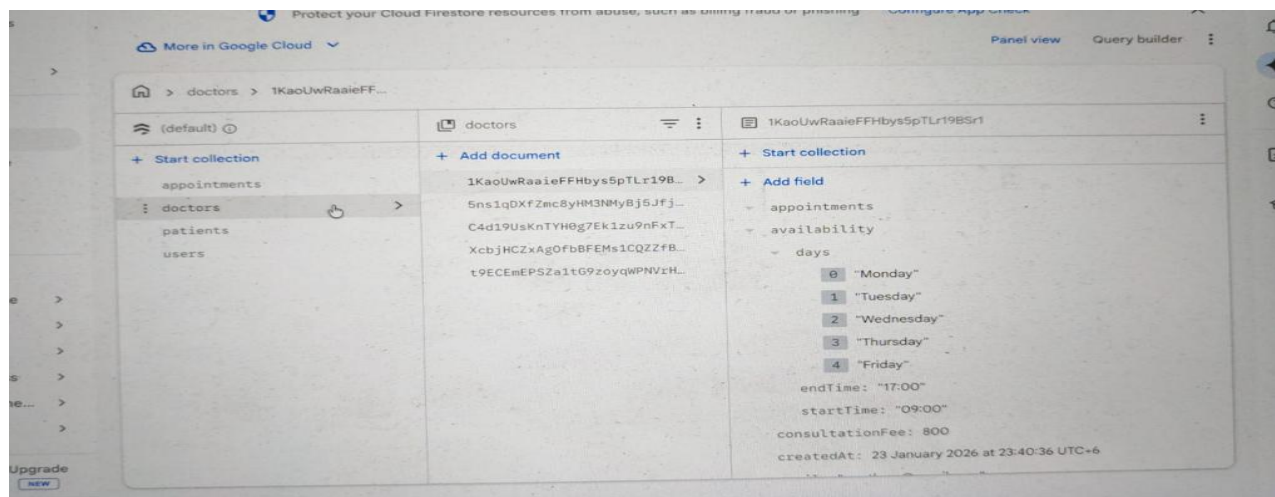
Figure : Logout

The Logout feature allows users patients, doctors, and administrators to securely exit the MediLink App after completing their tasks. By signing out, the user's session is terminated, preventing unauthorized access to sensitive information. This function is integrated with Firebase Authentication to ensure that all user sessions are properly managed. The logout process helps maintain data security, protects privacy, and ensures that the application remains safe for multi-user access on shared devices.

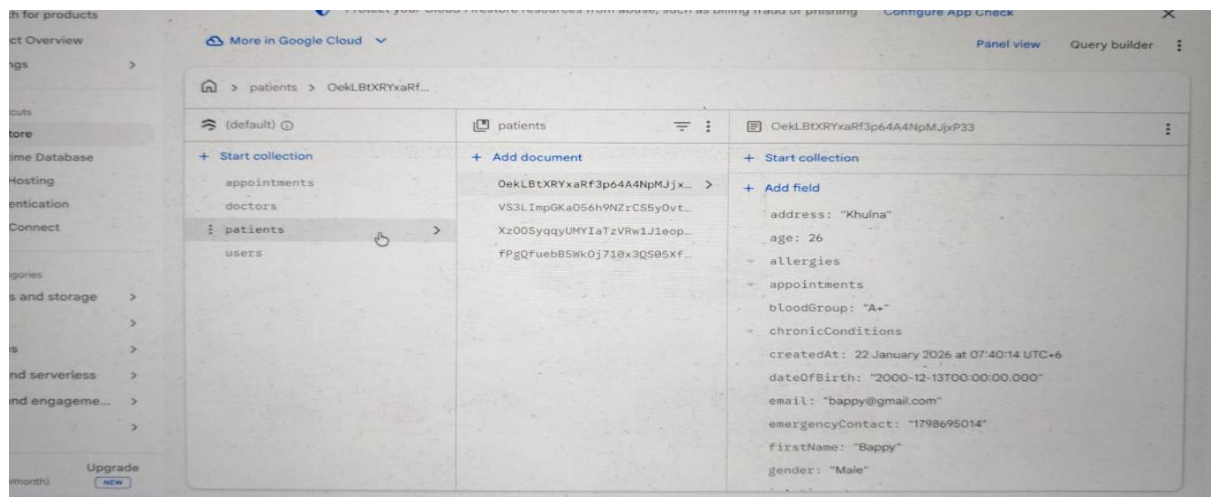
Database Appointments Tables



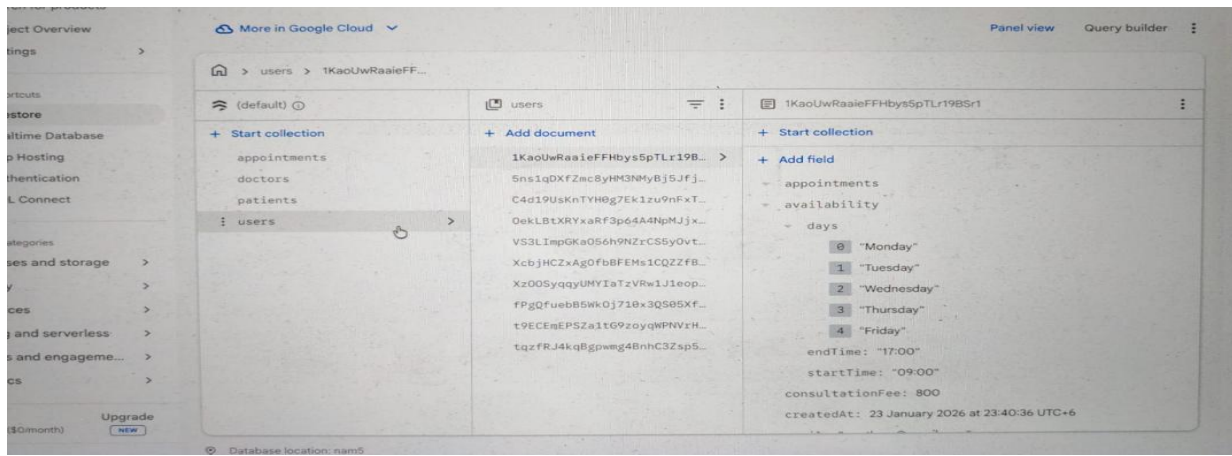
Database Doctors Tables



Database Patients Tables



Database Users Tables



The Medilink project database is designed with four main interconnected tables: Users, Doctors, Patients, and Appointments, which together manage the entire healthcare system efficiently. The Users table stores general information of all system users such as name, email, password, phone number, and role (admin, doctor, or patient). This table plays a central role in authentication, login system, and access control of the platform. The Doctors table contains detailed professional information about doctors including their specialization, qualifications, years of experience, chamber location, available time slots, and consultation fees, which helps patients easily find and select suitable doctors based on their needs. The Patients table stores personal and basic medical information of patients such as age, gender, blood group, address, and medical history, which assists doctors in understanding patient conditions and providing better treatment. Finally, the Appointments table manages the interaction between doctors and patients by storing appointment details such as appointment date, time, problem description, and status (pending, confirmed, completed, or canceled). This table ensures proper scheduling, tracking, and management of all medical appointments within the system, making the overall healthcare process organized, efficient, and user-friendly.